

Genetics and Epigenetics

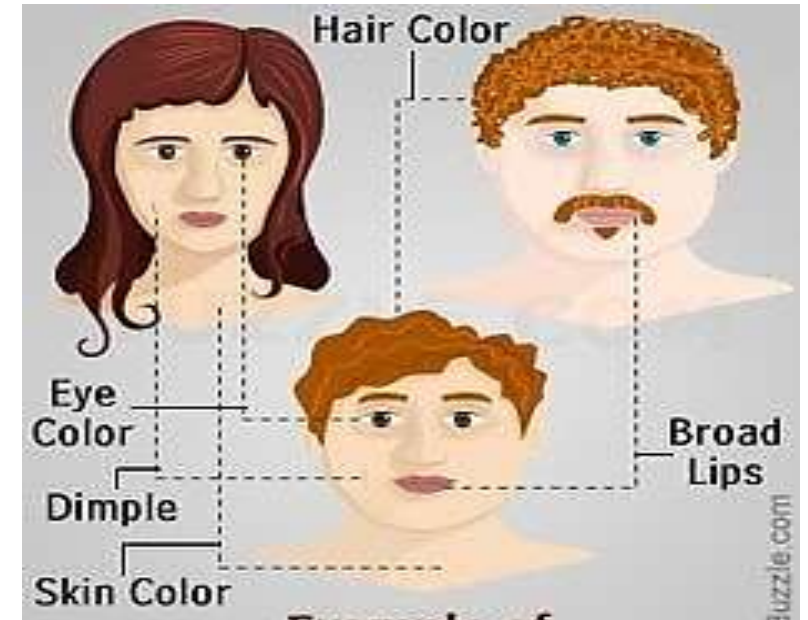
Introduction to Biology
Chapter: 7



Inspiring Excellence

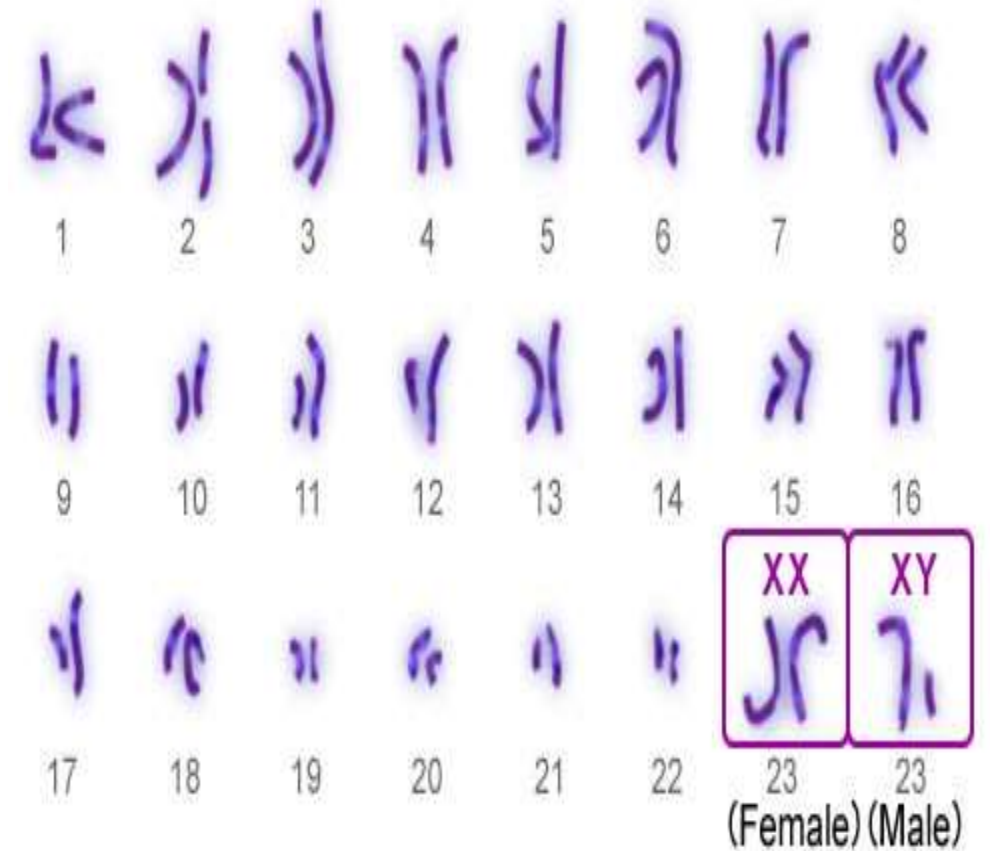
Have you ever wondered why children have similarities to both their parents?

- We acquire these similar characteristics (traits) through a process called **inheritance** (heredity).
- Inheritance occurs through transfer of genetic material from parents to their offspring.



Chromosomes

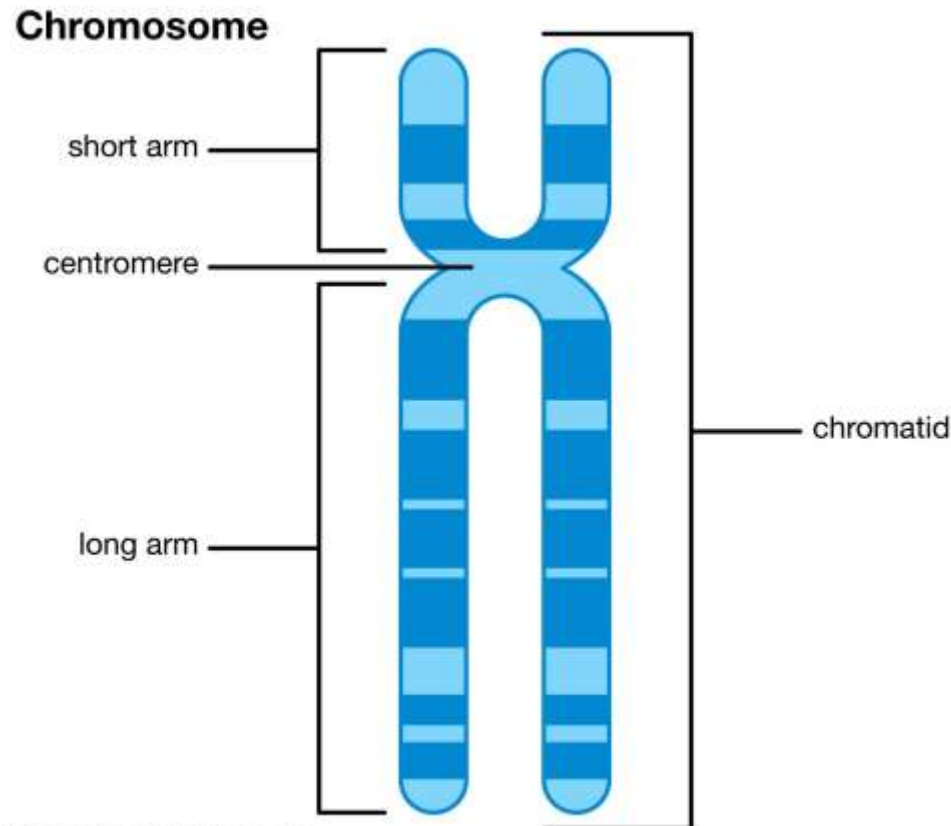
- They are present in the nucleus of the cell (eukaryotes) or present in the cytoplasm (prokaryotes).
- Different organisms have different numbers of chromosomes.
- Humans have 23 pairs of chromosomes (46 in total): 1 from each pair comes from father and one from mother.



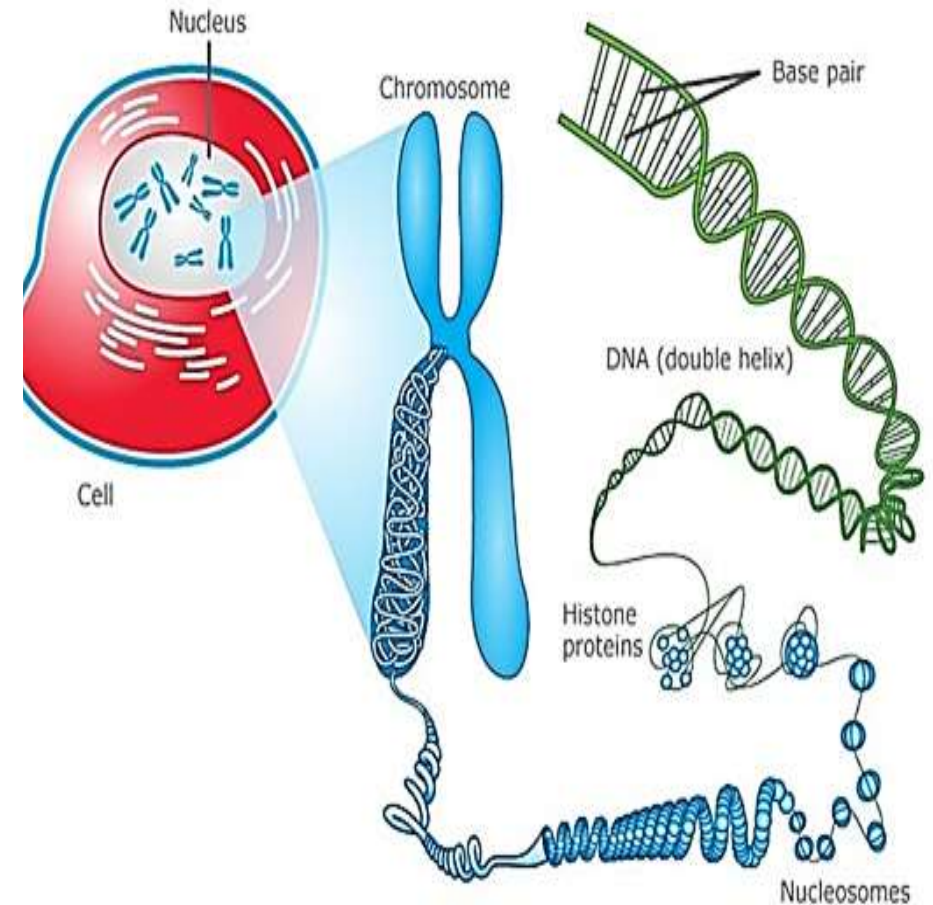
Karyotyping

Chromosomes

Chromosome: thread like structures that contain an organism's DNA. DNA is a long stretch of nucleotide polymer that is wrapped with histones (protein) and form nucleosomes. Nucleosomes pack tightly and ultimately fold organize into chromosomes.

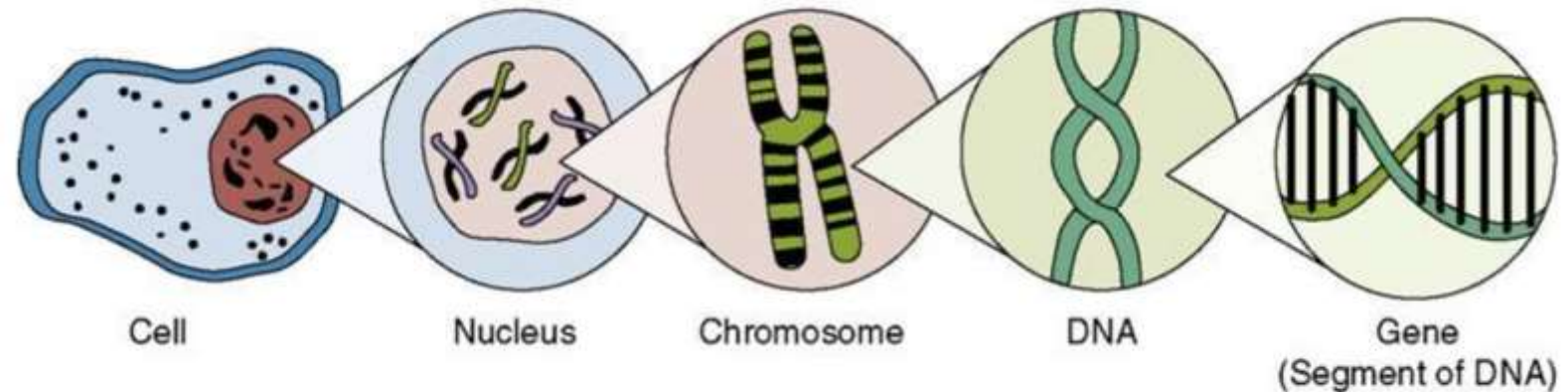
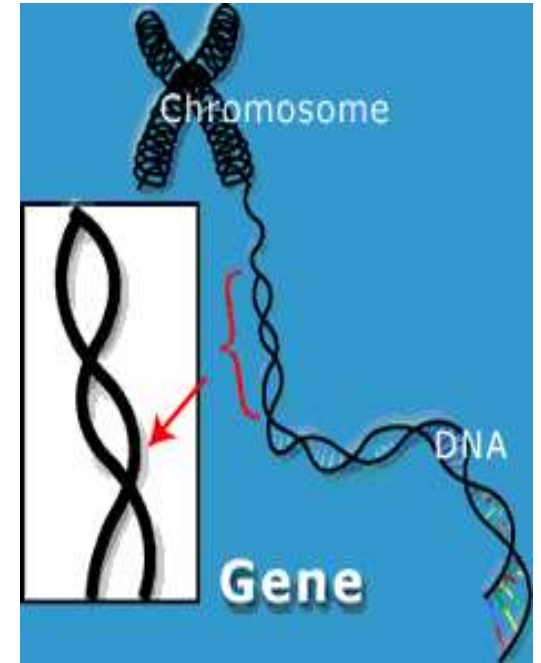


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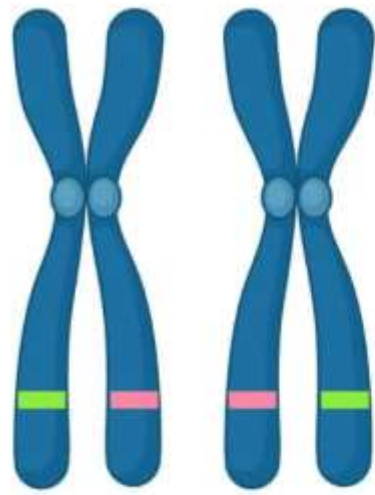
Genes

- Units of information about specific traits
- A sequence of DNA
- Passed from parents to offspring
- The complete set of DNA of an organism is called its **genome**.
- The study of genes, genetic variation and heredity is called **Genetics**.



Alleles

- Different molecular forms of the same gene
- Arise by mutation
- Each allele is passed from each parent to the offspring

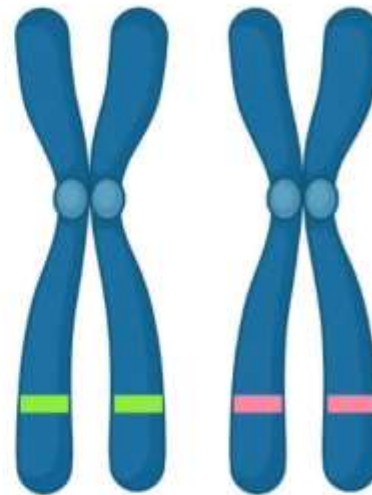


Aa

aA

Heterozygous Alleles

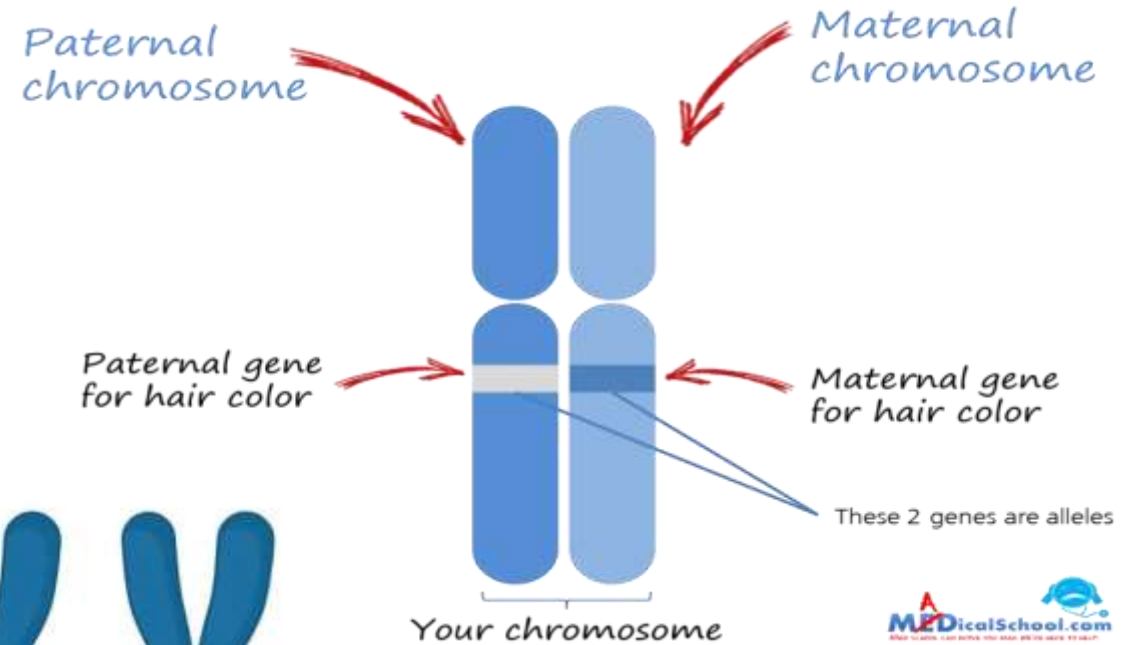
Alleles



AA

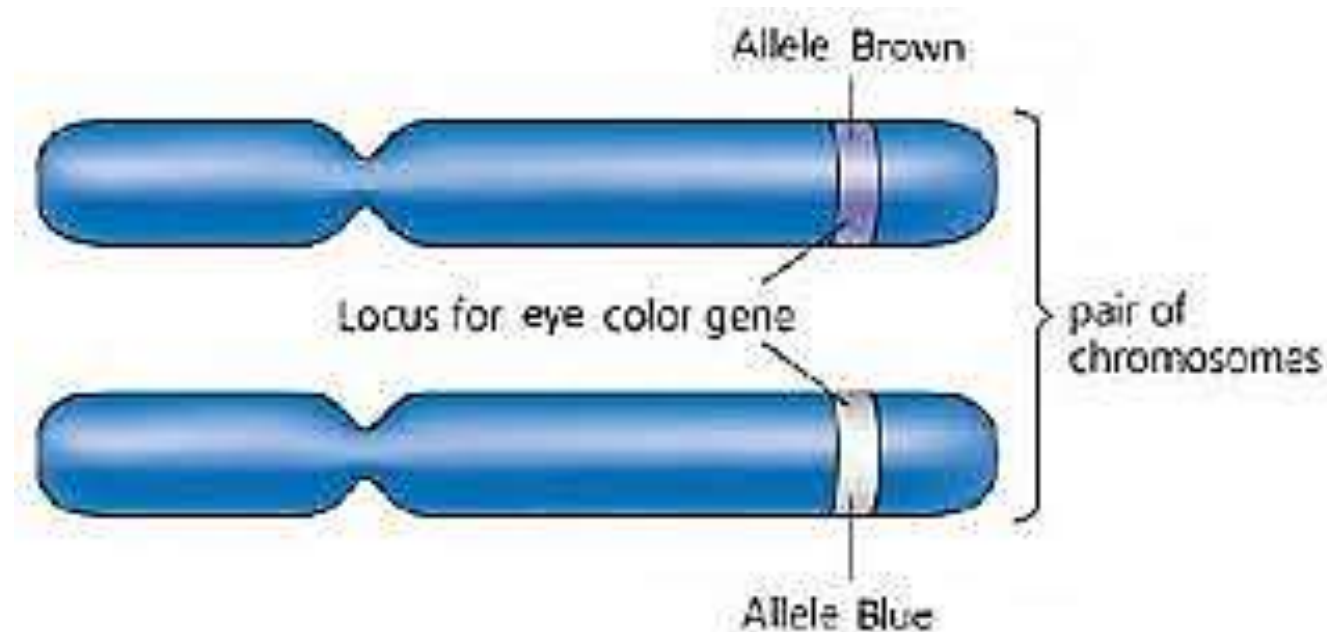
aa

Homozygous Alleles



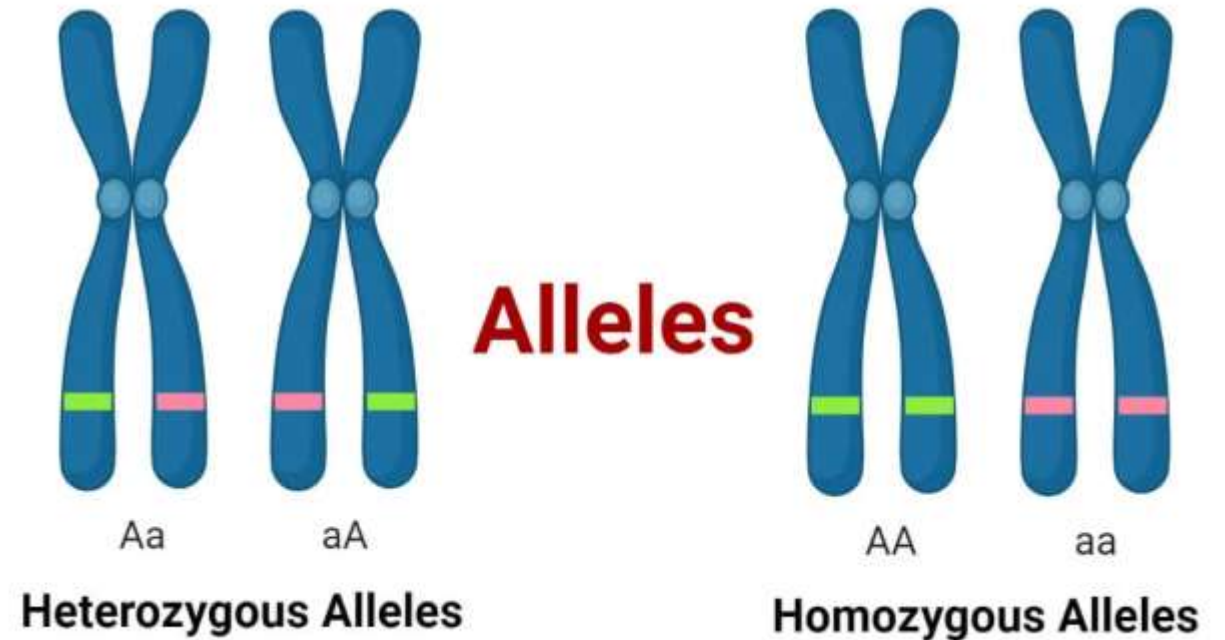
Locus

- Each gene has a specific location (locus) on a chromosome
- Alleles are located in the same locus of a chromosome



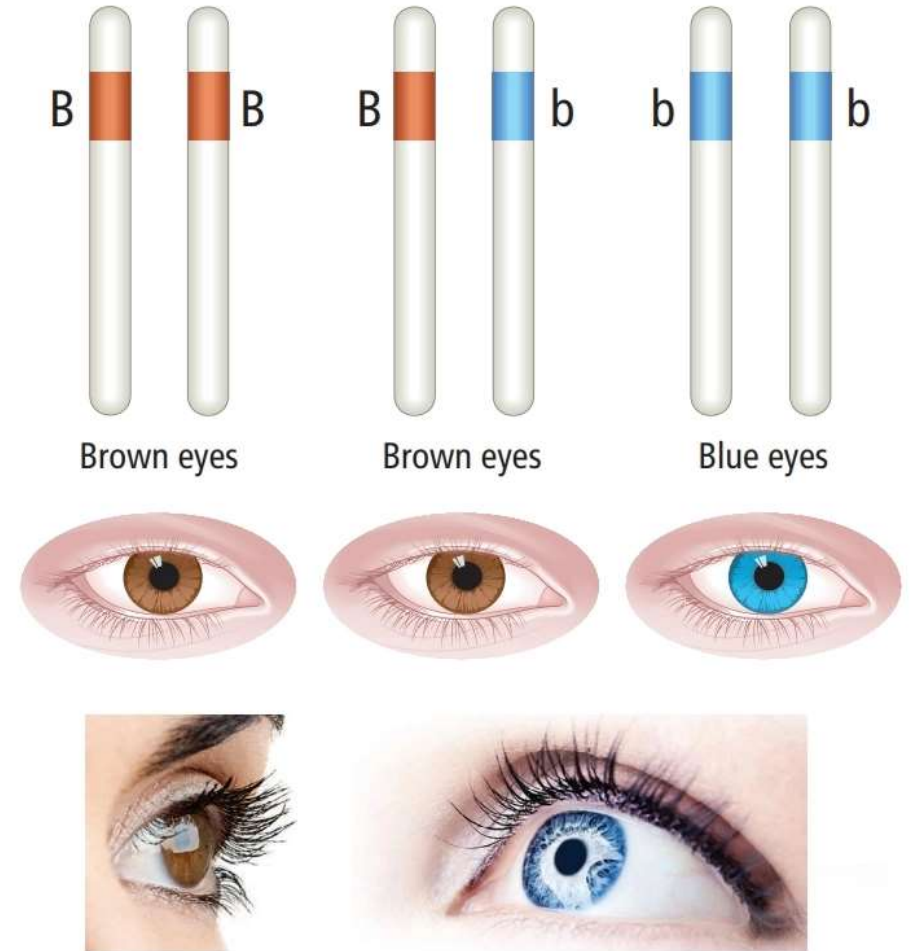
Allele combinations

- Homozygous
 - having two identical alleles at a locus
 - AA or aa
- Heterozygous
 - having two different alleles at a locus
 - Aa

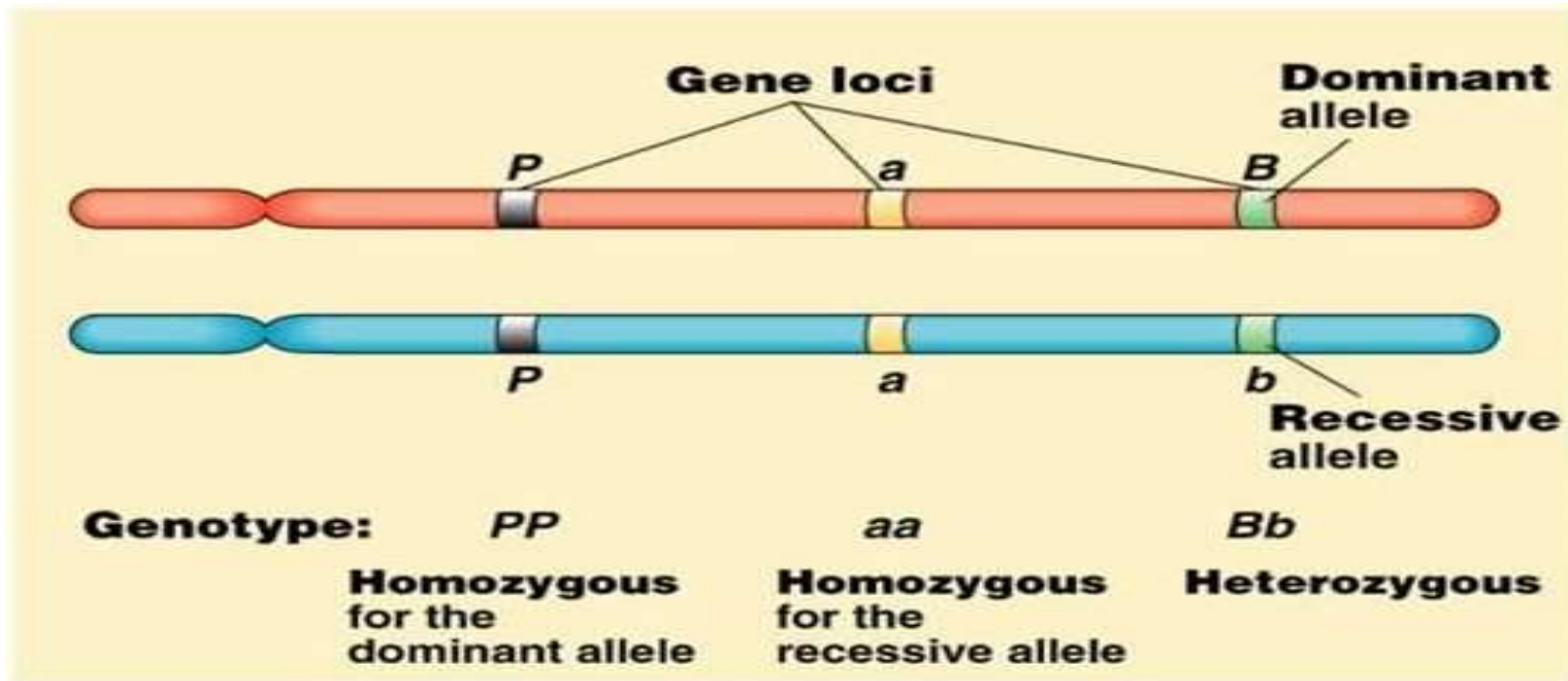


Dominant & Recessive Alleles

- A dominant allele is always expressed and hence masks a recessive allele that is paired with it.
- Recessive alleles are only expressed when both alleles are recessive.
- Dominant alleles are written in capital letters (B) and recessive alleles are written in small letters (b).



Examples



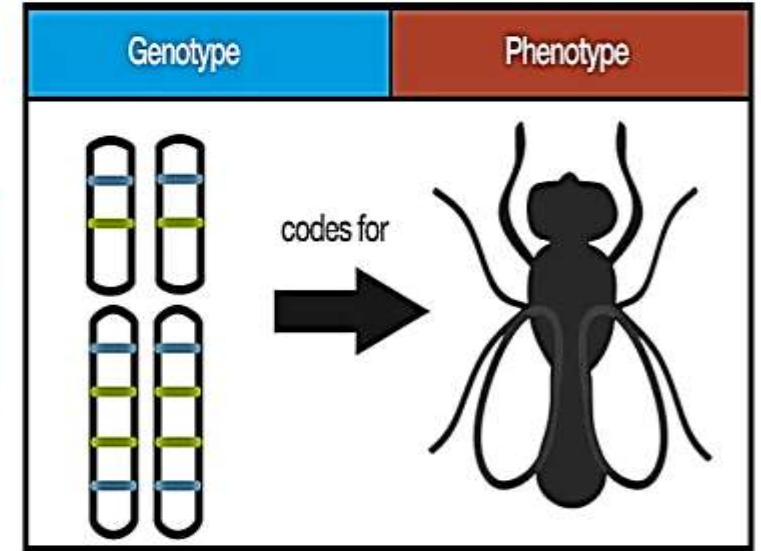
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Allele	Term	Result
1 dominant + 1 dominant	Homozygous dominant (BB)	Dominant trait
1 recessive + 1 recessive	Homozygous recessive (bb)	Recessive trait
1 dominant + 1 recessive	Heterozygous (Bb)	Dominant trait

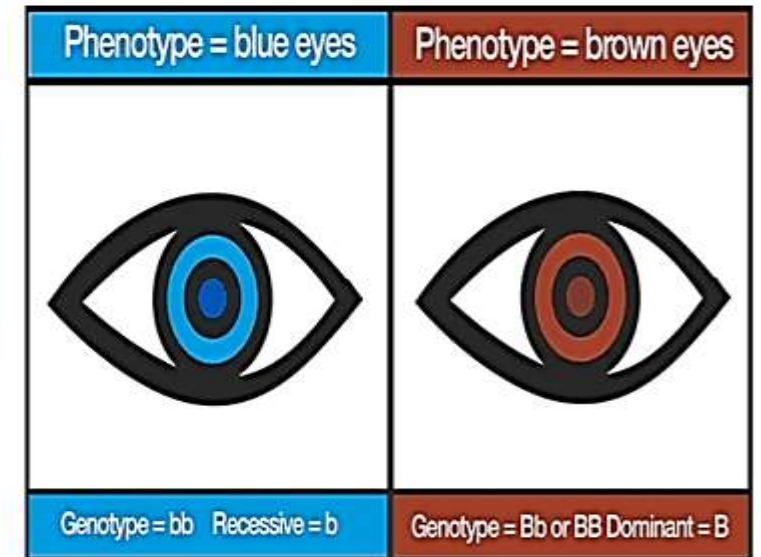
Genotype & Phenotype

- Genotype refers to particular genes an individual carries
- Phenotype refers to an individual's observable traits
- We cannot always determine genotype by observing phenotype:
Homozygous dominant or
Heterozygous?

Genotype is the genetic makeup of the organism



Phenotype is the physical appearance of the organism



GENOTYPE VERSUS PHENOTYPE

Genotype is the genetic makeup of an organism

Can be determined by observing DNA by genotyping methods

Completely depends on the gene sequences

Inherited by the offspring

Consists of all hereditary information that is the expressed and suppressed genes

Phenotype is the morphology, properties and behavior of an organism

Can be determined by observing outward characters

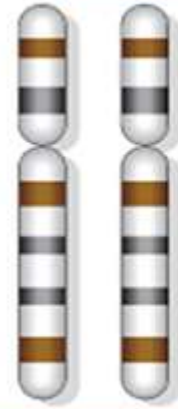
Depends on the genotype and environmental factors

Not inherited by the offspring

Consists of expressed genes

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Genotype Vs. Phenotype



GENOTYPE



PHENOTYPE

Gregor Mendel

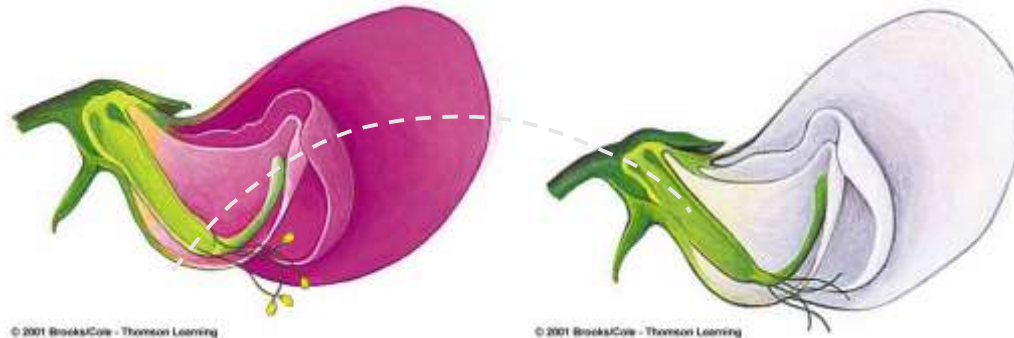
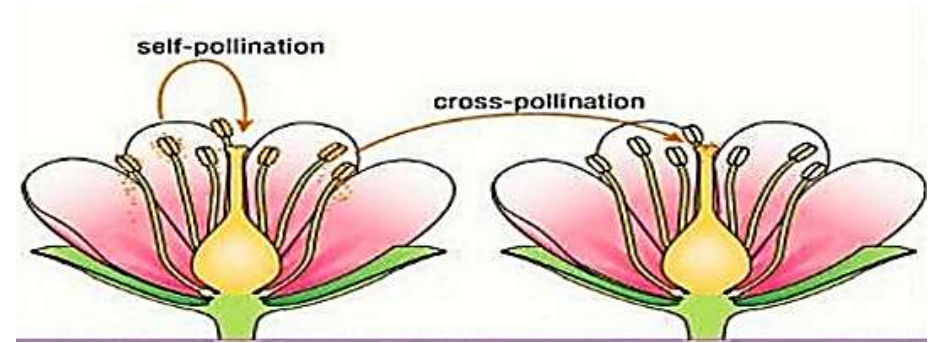
- Strong background in plant breeding and mathematics.
- Using pea plants, found indirect but observable evidence of how parents transmit genes to offspring.
- Particulate theory: physical traits are inherited as 'particles'



Father of genetics!

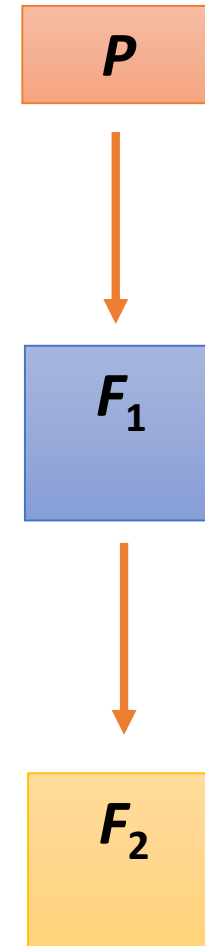
The Garden Pea Plant

- Self-pollinating
- True breeding (different alleles not normally introduced)
- Can be experimentally cross-pollinated
- Easy to cultivate
- Shorter lifespan



Tracking Generations

- Parental generation mates to produce:
- First-generation offspring mate to produce:
- Second-generation offspring



Color of Pea plants: Monohybrid cross

- A mix of genetics between two homozygous genotypes:
 - Either completely dominant or completely recessive
- Mono: one
- Hybrid: combination of two organisms
- For garden pea plants, Mendel observed the colour of pea plants

Female gametes

a

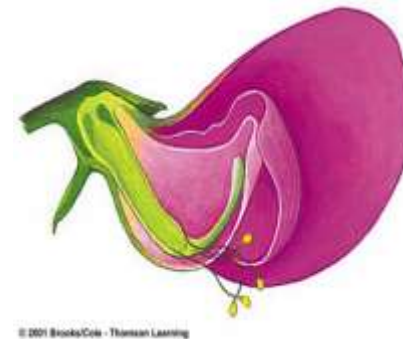
a

	<i>a</i>	<i>a</i>
<i>A</i>	<i>Aa</i>	<i>Aa</i>
<i>A</i>	<i>Aa</i>	<i>Aa</i>

Male gametes

A

A



Purple: AA



White: aa

F_1 Results of One Monohybrid Cross

Result

All offspring were
heterozygous dominant
(Aa)

True-breeding
homozygous recessive
parent plant



aa

True-breeding
homozygous dominant
parent plant



AA

A

A

a a

Aa	Aa
Aa	Aa



F_1
PHENOTYPES



Aa



Aa



Aa

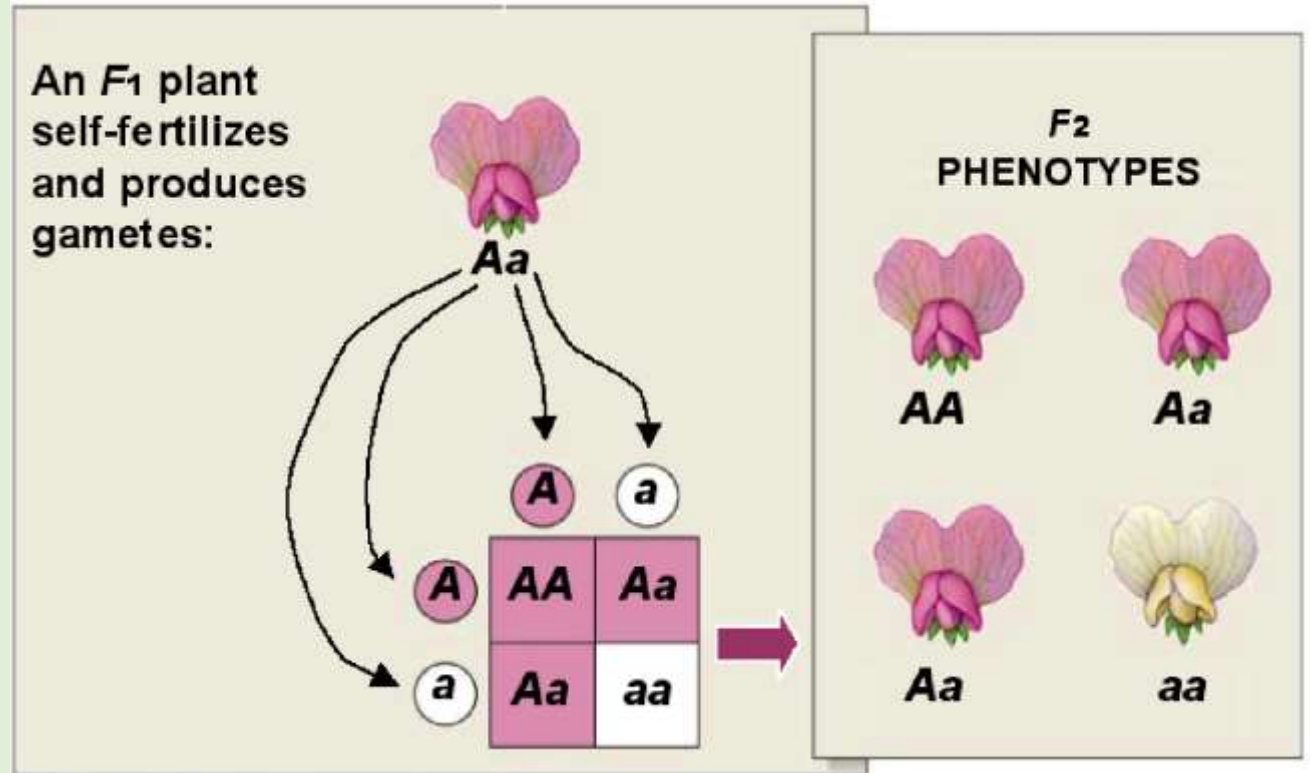


Aa

F_2 Results of Monohybrid Cross

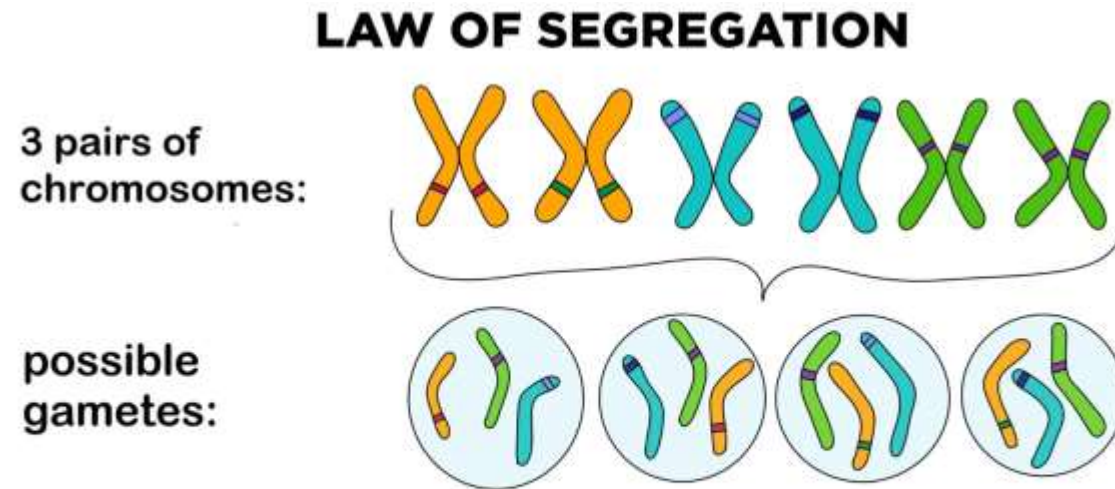
Result

- 2 offspring were **heterozygous dominant (Aa)**
- 1 offspring was **homozygous dominant (AA)**
- 1 offspring was **homozygous recessive (aa)**
- Phenotypic ratio – 3 dominant : 1 recessive



Mendel's Theory of Segregation

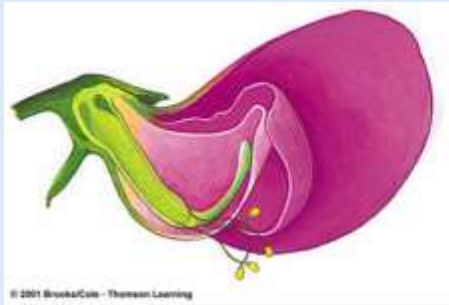
- An individual inherits a unit of information (allele) about a trait from each parent.
- During gamete formation, the alleles segregate from each other.



The 2 copies of each gene separate and end up in different gametes

Color and Stem of Pea Plants: Dihybrid Cross

- Experimental cross between individuals that are homozygous for different versions of **two** traits.
- Di = two
- Hybrid = combination of two organisms



Purple: AA



White: aa

+



Tall: BB



Dwarf: bb

F_1 Results of Mendel's Dihybrid Crosses

$AABB$ x $aabb$

	ab	ab	ab	ab
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$
AB	$AaBb$	$AaBb$	$AaBb$	$AaBb$

- All plants displays the dominant form of both traits (Purple and tall)
- We now know:
 - All plants inherit one allele for each trait from each parent
 - All plants were heterozygous ($AaBb$)

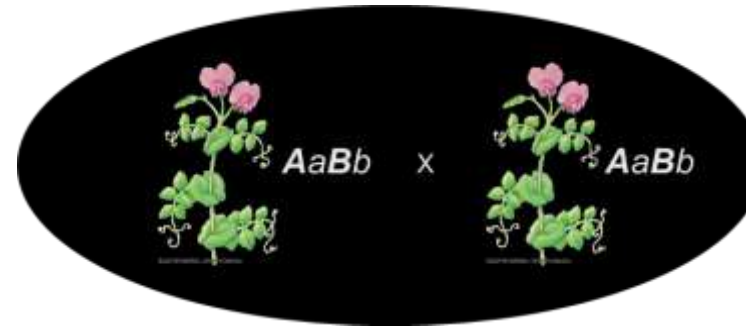
Phenotypic Ratios in F_2

If the two traits are coded for by genes on separate chromosomes, sixteen allele combinations are possible:

Four Phenotypes:

- Tall, purple-flowered (9/16)
- Tall, white-flowered (3/16)
- Dwarf, purple-flowered (3/16)
- Dwarf, white-flowered (1/16)

Phenotypic ratio: 9:3:3:1



	$\frac{1}{4}$ AB	$\frac{1}{4}$ Ab	$\frac{1}{4}$ aB	$\frac{1}{4}$ ab
$\frac{1}{4}$ AB	$\frac{1}{16}$ $AABB$	$\frac{1}{16}$ $AABb$	$\frac{1}{16}$ $AaBB$	$\frac{1}{16}$ $AaBb$
$\frac{1}{4}$ Ab	$\frac{1}{16}$ $AABb$	$\frac{1}{16}$ $AAbb$	$\frac{1}{16}$ $AaBb$	$\frac{1}{16}$ $Aabb$
$\frac{1}{4}$ aB	$\frac{1}{16}$ $AaBB$	$\frac{1}{16}$ $AaBb$	$\frac{1}{16}$ $aaBB$	$\frac{1}{16}$ $aaBb$
$\frac{1}{4}$ ab	$\frac{1}{16}$ $AaBb$	$\frac{1}{16}$ $Aabb$	$\frac{1}{16}$ $aaBb$	$\frac{1}{16}$ $aabb$

Mendel's Law of Independent Assortment

- Mendel concluded that the two “units” for the first trait were to be sorted into gametes independently of the two “units” for the other trait.
- Members of each pair of homologous chromosomes are sorted into gametes at random during meiosis.

MENDEL'S FIRST LAW
VERSUS
MENDEL'S SECOND LAW

MENDEL'S FIRST LAW

A principle that describes the separation of the two copies of each hereditary factor during the production of gametes

Also called the law of segregation

Uses a monohybrid cross

Ratio of offspring is 3:1

MENDEL'S SECOND LAW

A principle that describes the independent assortment of alleles of different genes during the formation of gametes

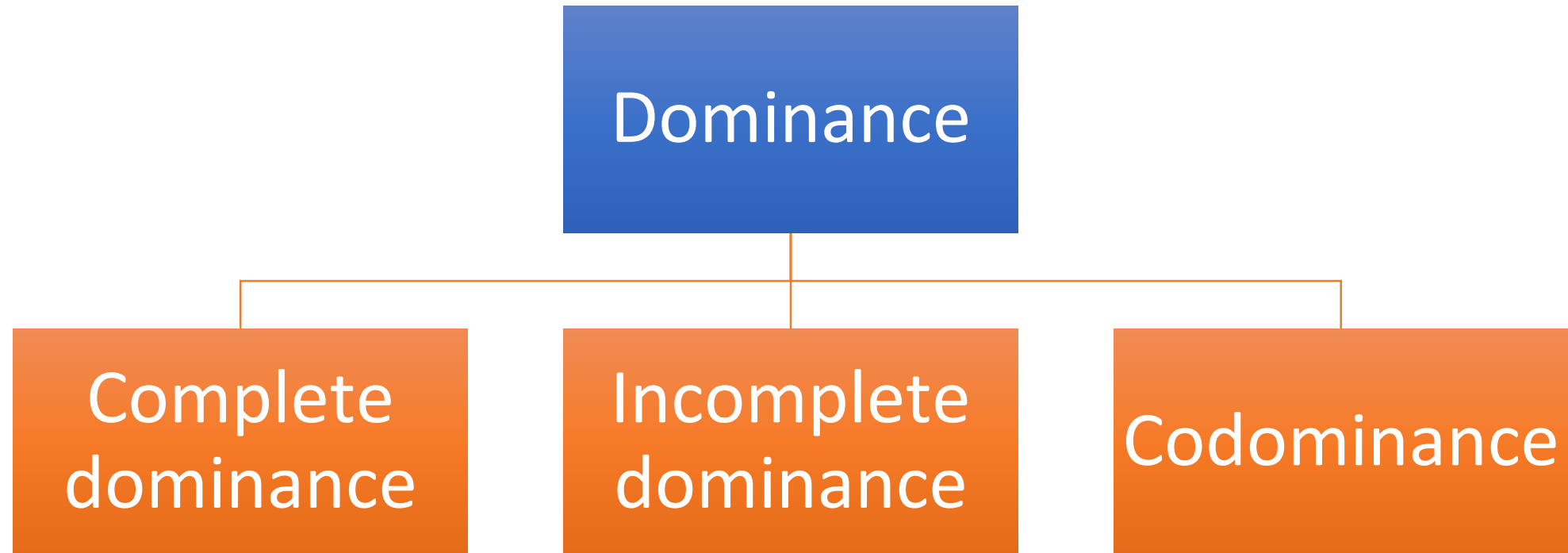
Also called the law of independent assortment

Uses a dihybrid cross

Ratio of offspring is 9:3:3:1

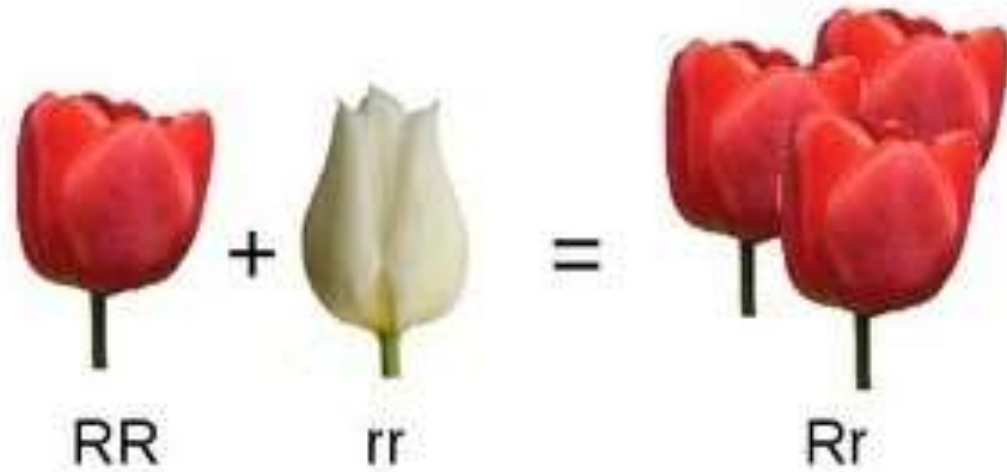
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Dominance Relations



Complete dominance

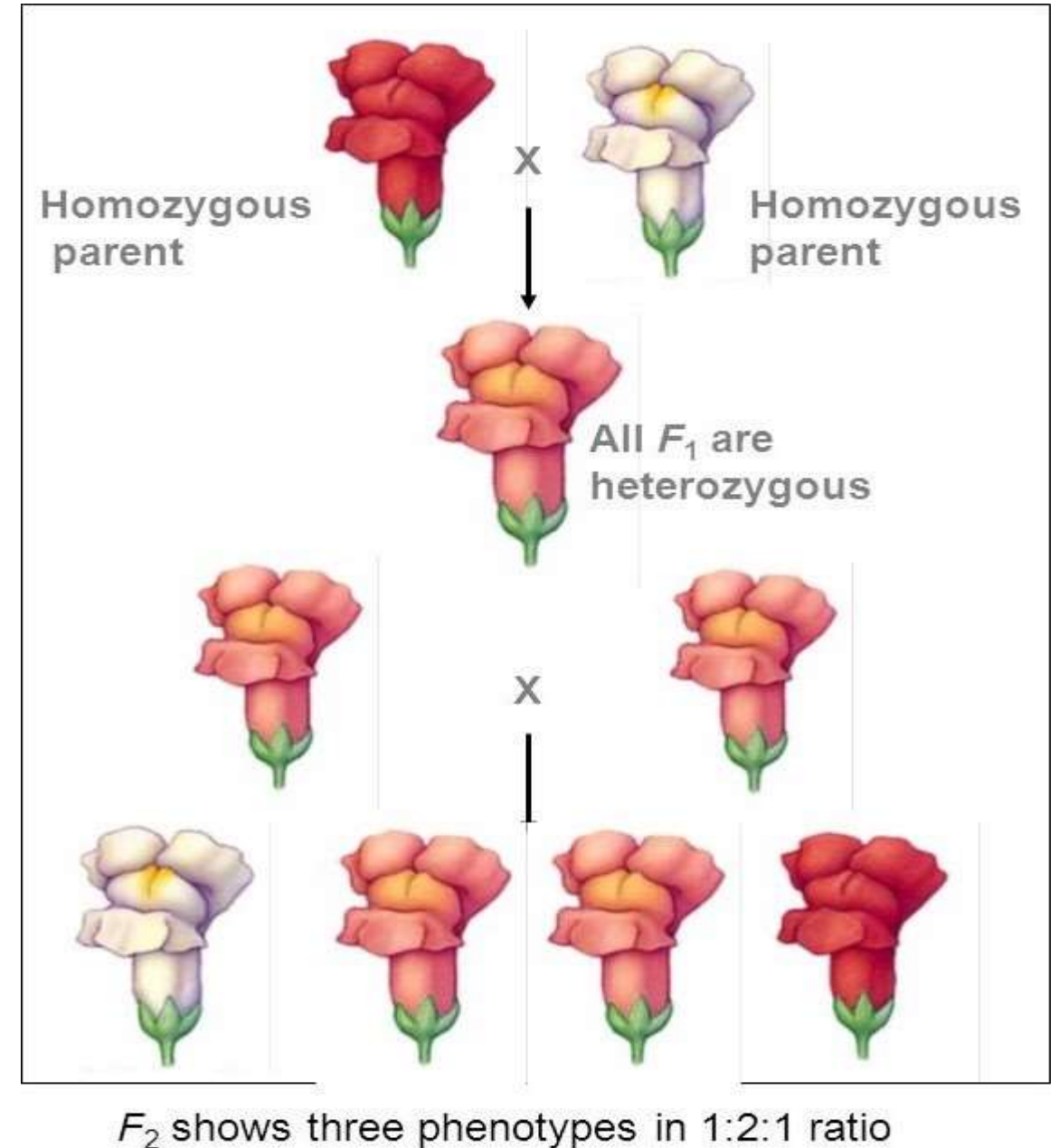
- Dominance in heterozygous condition where the dominant allele completely masks the effects of the recessive allele.



Incomplete dominance

- Dominant allele does not completely mask the effect of the recessive allele.
- Heterozygous phenotype is somewhere between that of two homozygotes.
- Blend of both traits

Flower colour in Snapdragons



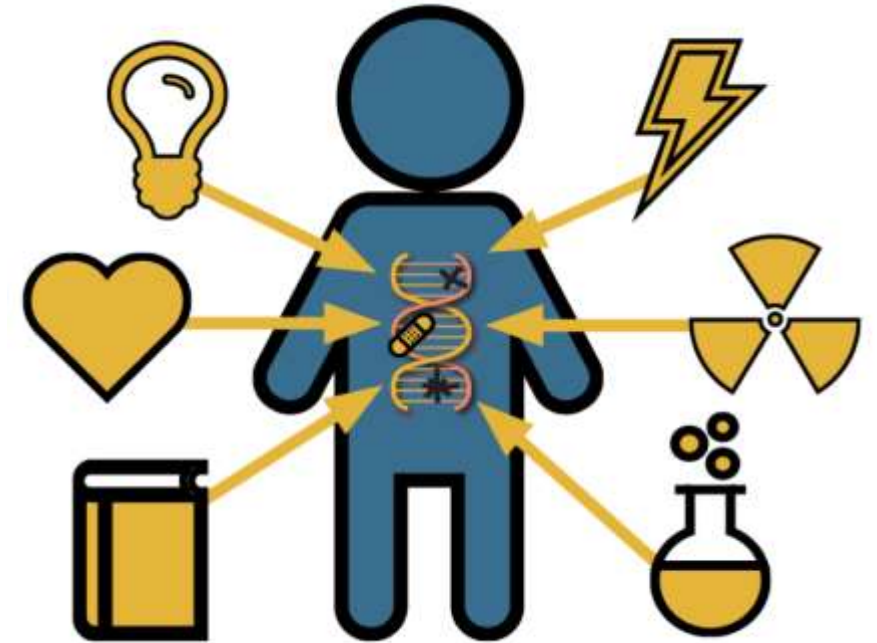
Codominance

- Both alleles of a particular trait are fully expressed.
- Non-identical alleles specify two phenotypes that are both expressed in heterozygotes.
- Offspring has a phenotype that shows combination of both traits.



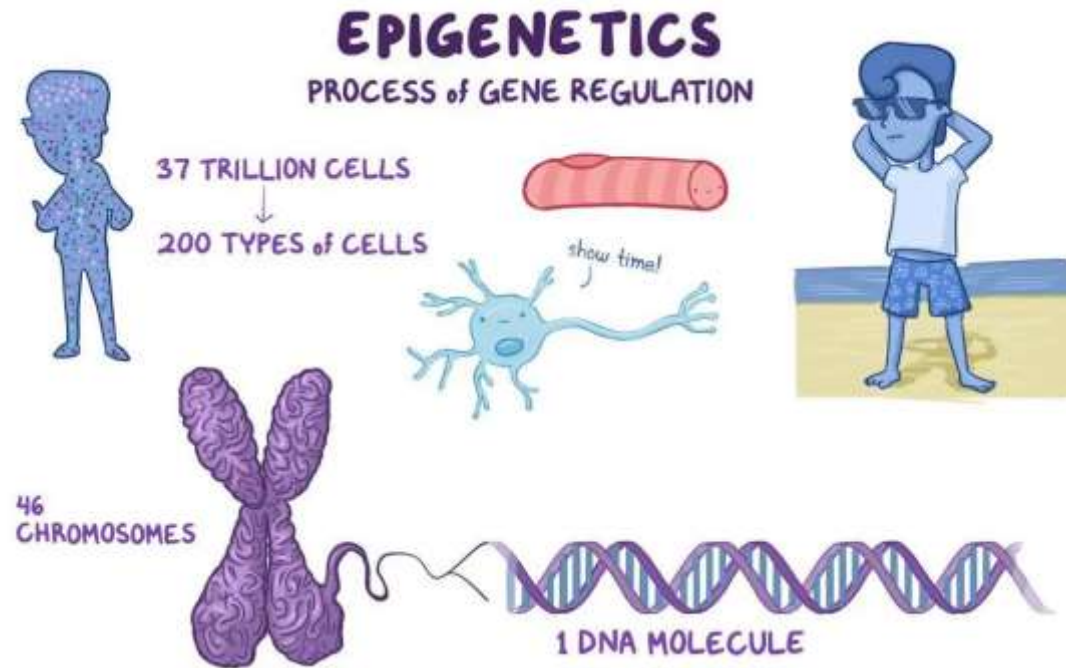
Epigenetics

- Epigenetics is the genetic control of a living organism through factors other than the DNA sequence
- During development, DNA accumulates chemical marks that affect how much of a gene is expressed. These chemical marks are called epigenomes
- Epigenetic changes can also switch certain genes 'on' or 'off'!



How Epigenetics Works?

- All our cells have the same DNA, so why do we even have different cells?



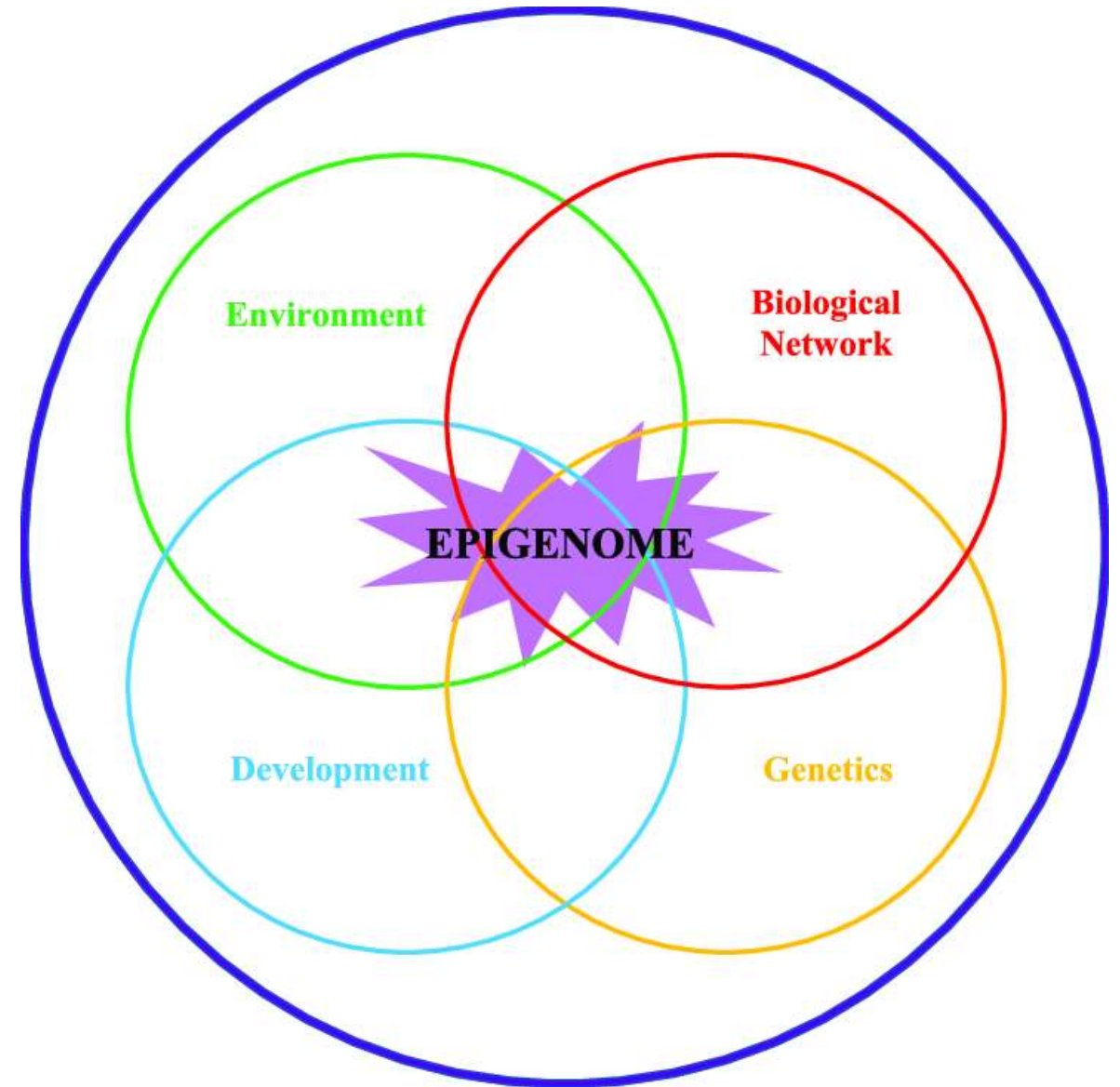
- Why are genotypically identical twins *not* phenotypically identical?

- What are the factors that cause people to develop differently?

Factors Affecting Epigenomes

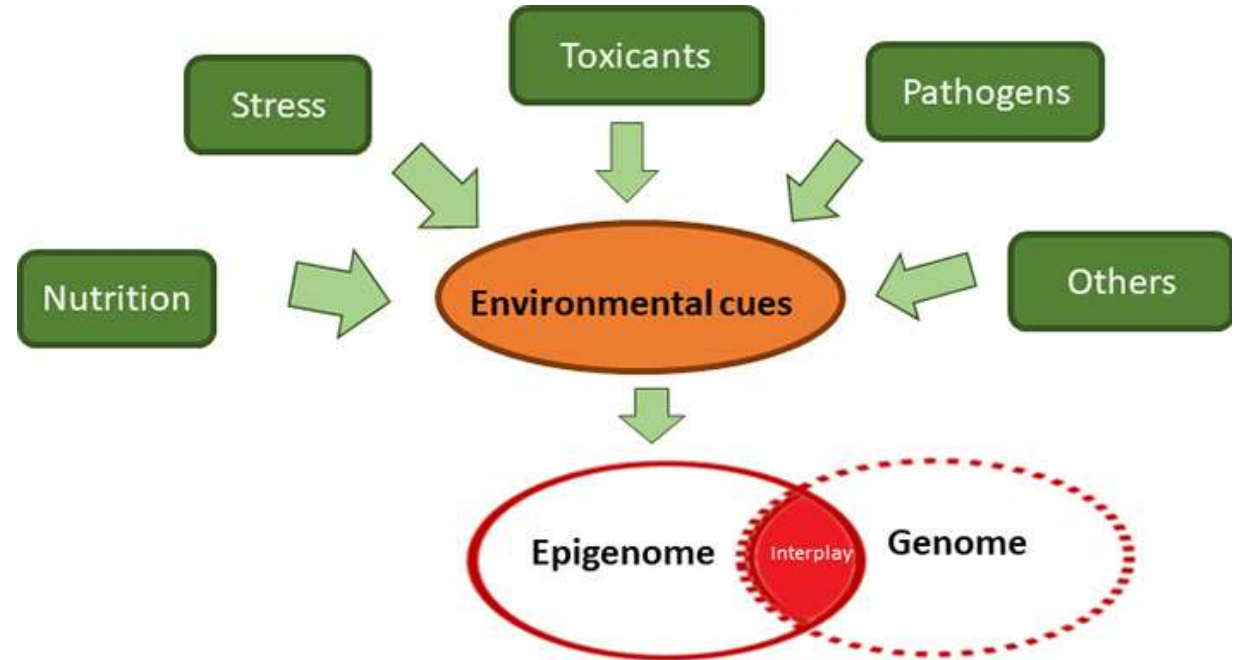
Four key factors:

1. Genetics
2. Environment
3. Biological Networks
4. Development



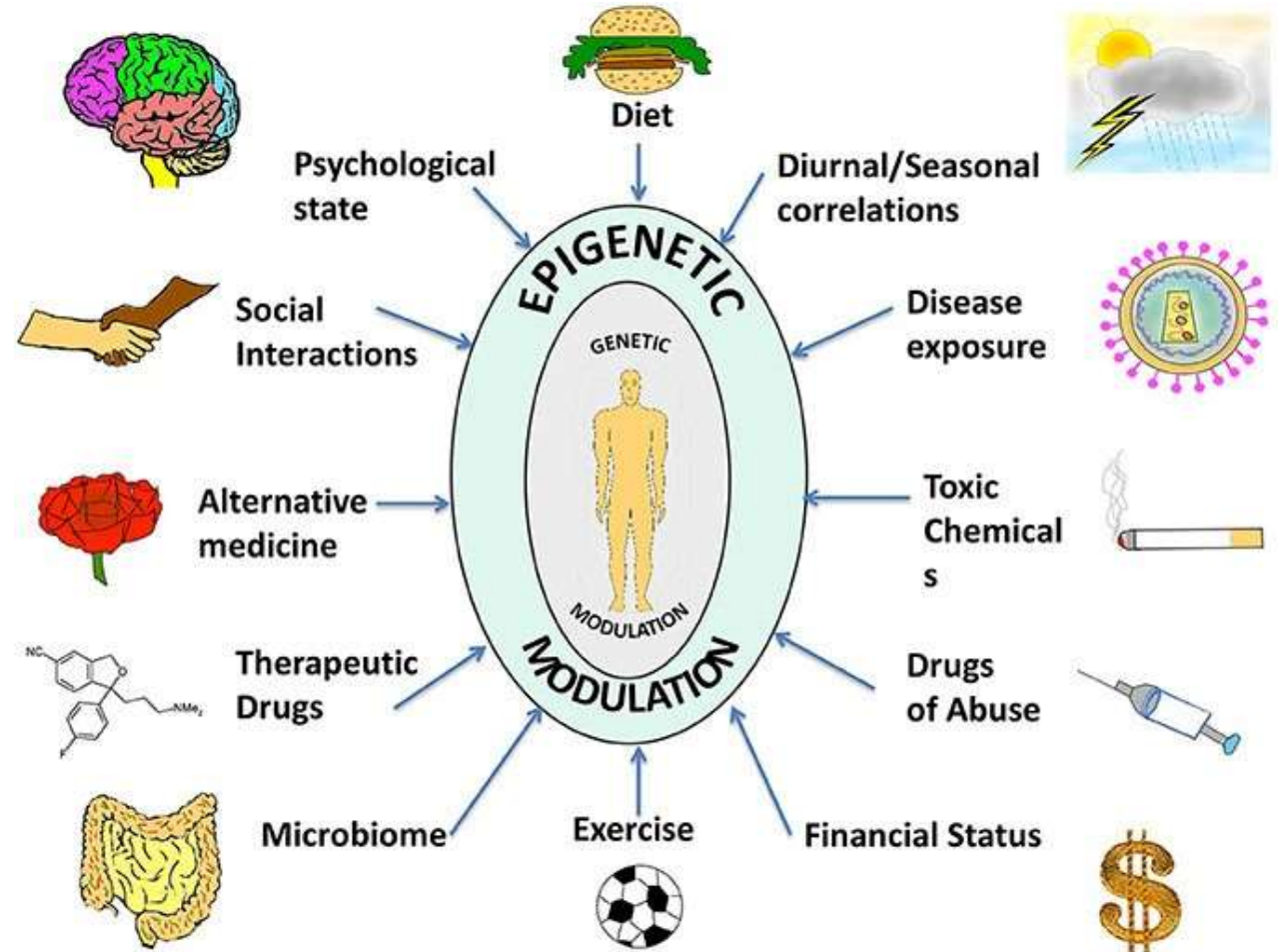
Genetics and Environment

- The protein structure and arrangement of genes themselves can sometimes influence how the chemical marks are distributed across the DNA
- Environmental factors consist of 'outside' factors that can have an impact on genes. These can be:
 - Pollutants
 - Allergens
 - Heavy metals
 - Chemical toxins
 - Electromagnetic radiation



Developmental Factors

- Development can include all stages of life, starting from embryonic, to early childhood, all the way through to adulthood
- Examples of developmental factors are:
 - Nutrient intake (balance of diet)
 - Stress and psychology
 - Smoking and alcohol
 - Social behavior patterns



Case study – Identical Twins, Different behaviour?

- Suppose there is a pair of identical twins – they share the same genome, and thus have the same genotype
- However, one twin likes to read books and spend time on the computer growing up, while the other likes to play musical instruments and prefers to spend more time outdoors.
- How is this difference explained through epigenetics?

